

System Capacity & Care Load Analytics for Unaccompanied Children

Research Paper

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Tool: Power BI

Abstract

This research explains a Power BI dashboard built from the HHS Unaccompanied Children dataset. The dashboard monitors children in CBP custody, HHS care, admissions, discharges, system load, backlog and monthly trends to support operational decisions.

Introduction

Government agencies need real-time analytics to understand capacity and care load. The dashboard converts raw operational data into interactive KPIs and visualizations for faster decision making.

Objectives

1. Monitor total system load.
2. Measure net intake.
3. Identify backlog.
4. Compare CBP custody and HHS care.
5. Support monthly trend analysis.

Dataset

The dashboard uses cleaned operational records with Date, Children in CBP Custody, Children in HHS Care, Children Apprehended, Children Discharged, Total System Load, Net Intake and Backlog.

Methodology

Data was cleaned in Power Query, calculated columns were created, KPIs were generated, and interactive visuals were designed in Power BI.

Dashboard Explanation

KPI Cards summarize Total System Load, HHS Care, CBP Custody and Backlog. A line chart shows monthly trends. A clustered bar chart compares CBP and HHS values. A waterfall/column visual explains Net Intake. Slicers filter by month and year.

Key Findings

Higher admissions increase system load. When discharges are lower than intake, backlog grows. Trend visuals help identify periods requiring additional care capacity.

Conclusion

The dashboard demonstrates how Power BI can transform public health and child welfare data into actionable insights for planning, resource allocation and performance monitoring.

Future Scope

Add predictive forecasting, regional comparison, demographic analysis and live data integration.